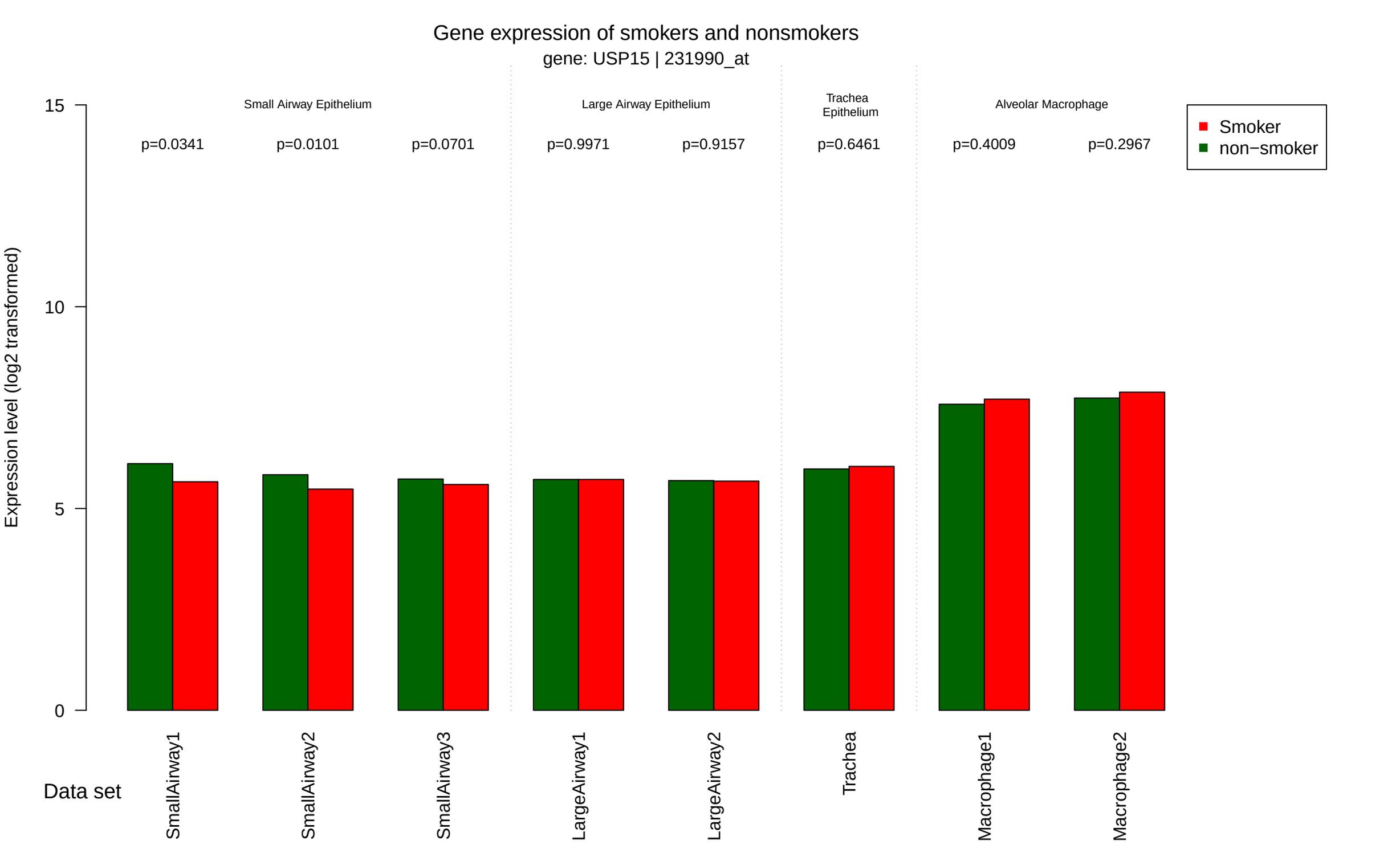
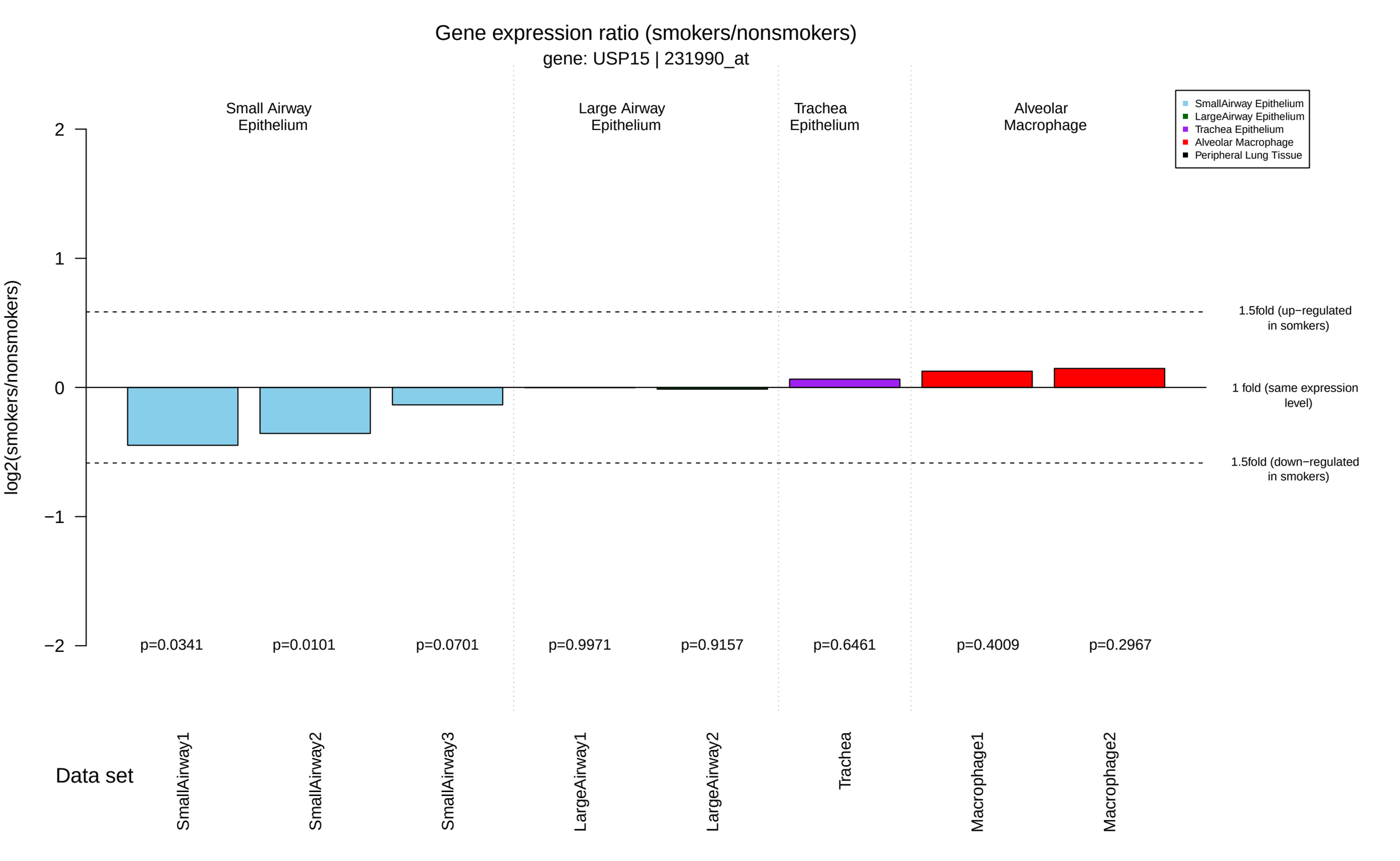
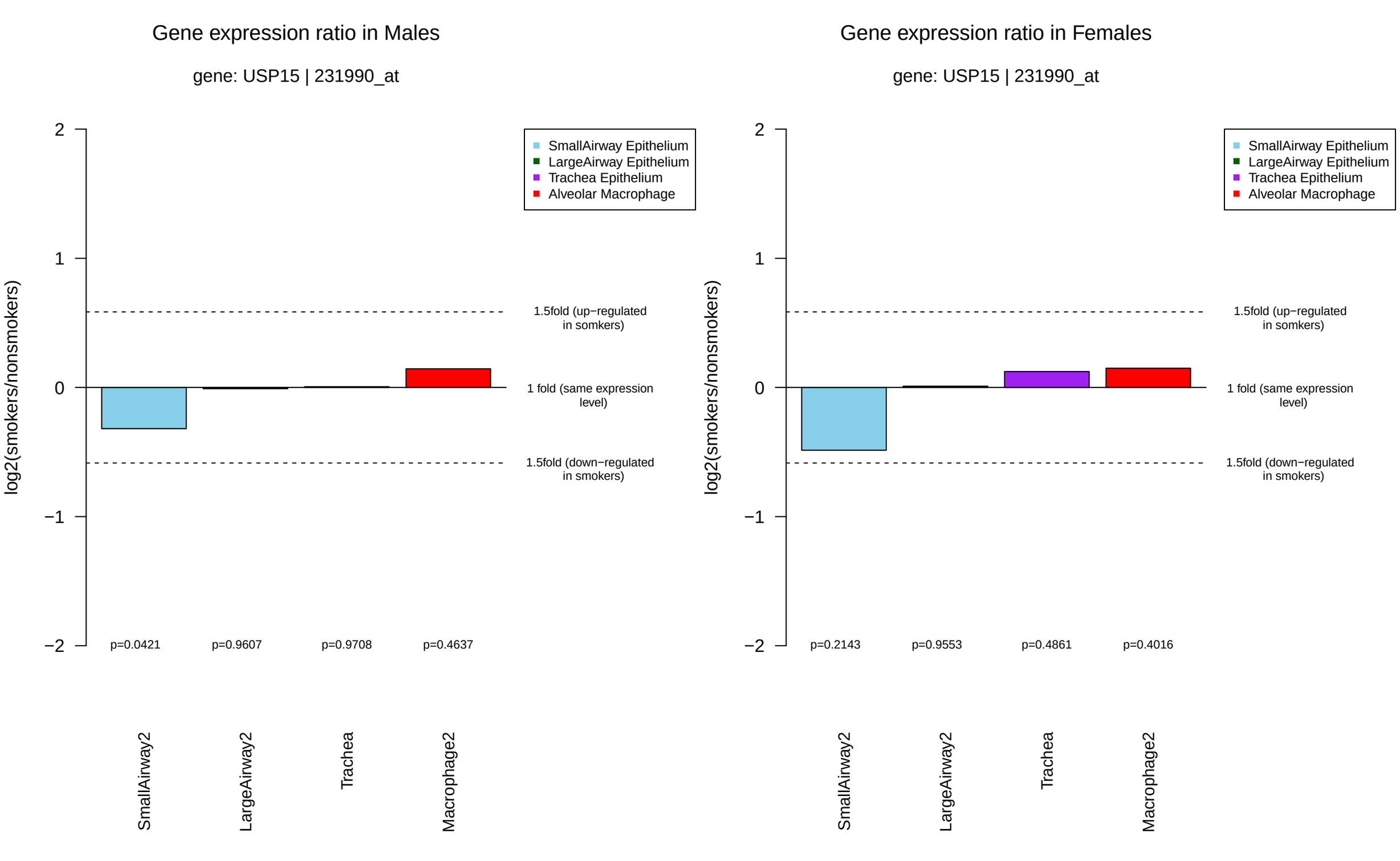
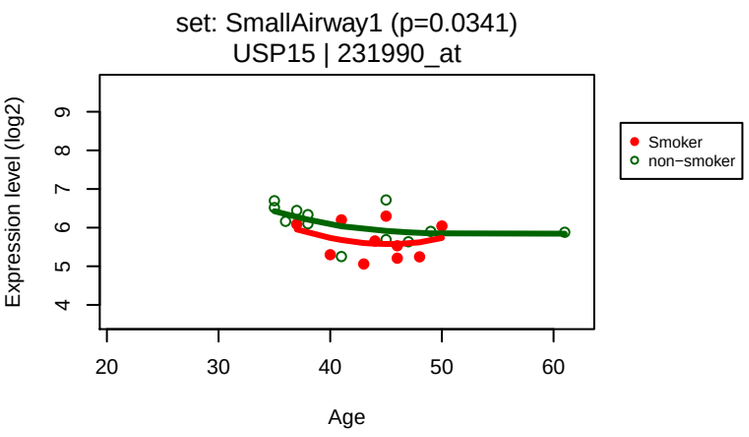
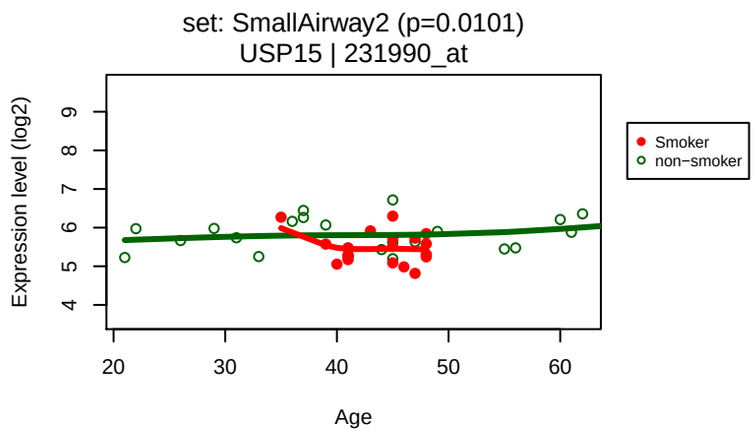
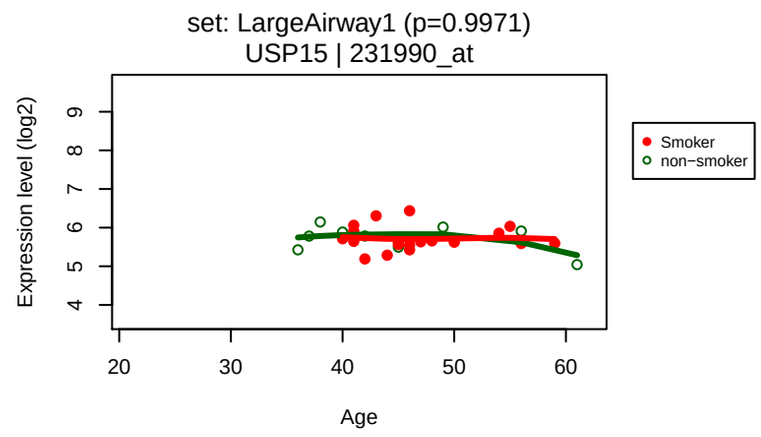

Gene expression of smokers and nonsmokers
gene: USP15 | 231990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0341
p=0.0101
p=0.0701
p=0.9971
p=0.9157
p=0.6461
p=0.4009
p=0.2967
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

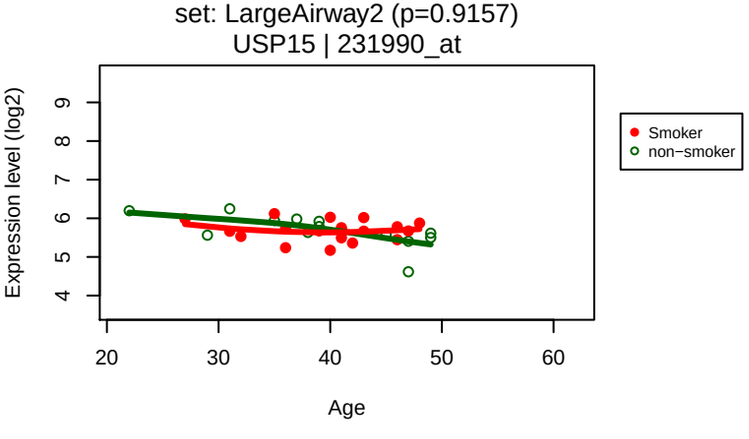

Gene expression ratio (smokers/nonsmokers)
gene: USP15 | 231990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0341
p=0.0101
p=0.0701
p=0.9971
p=0.9157
p=0.6461
p=0.4009
p=0.2967
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

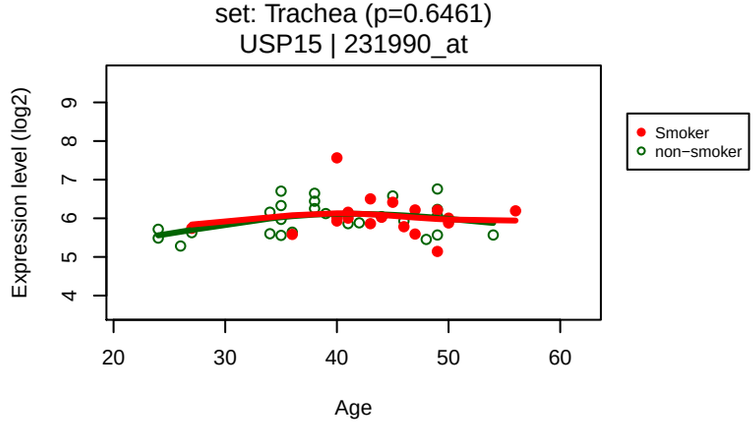

Gene expression ratio in Males
gene: USP15 | 231990_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0421
p=0.9607
p=0.9708
p=0.4637
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP15 | 231990_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2143
p=0.9553
p=0.4861
p=0.4016
SmallAirway2
LargeAirway2
Trachea
Macrophage2

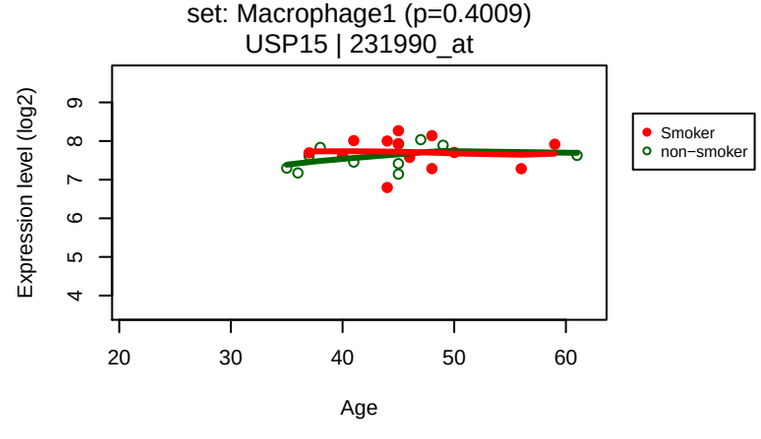

set: SmallAirway1 (p=0.0341)
USP15 | 231990_at
Expression level (log2)
Age
Smoker
non−smoker

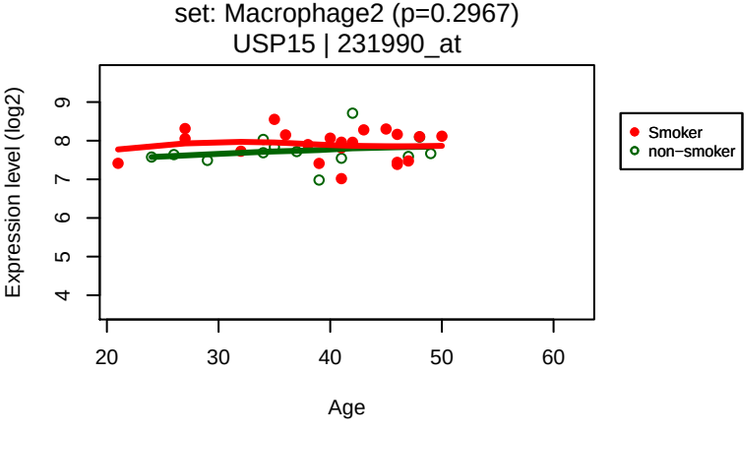

set: SmallAirway2 (p=0.0101)
USP15 | 231990_at
Expression level (log2)
Age
Smoker
non−smoker

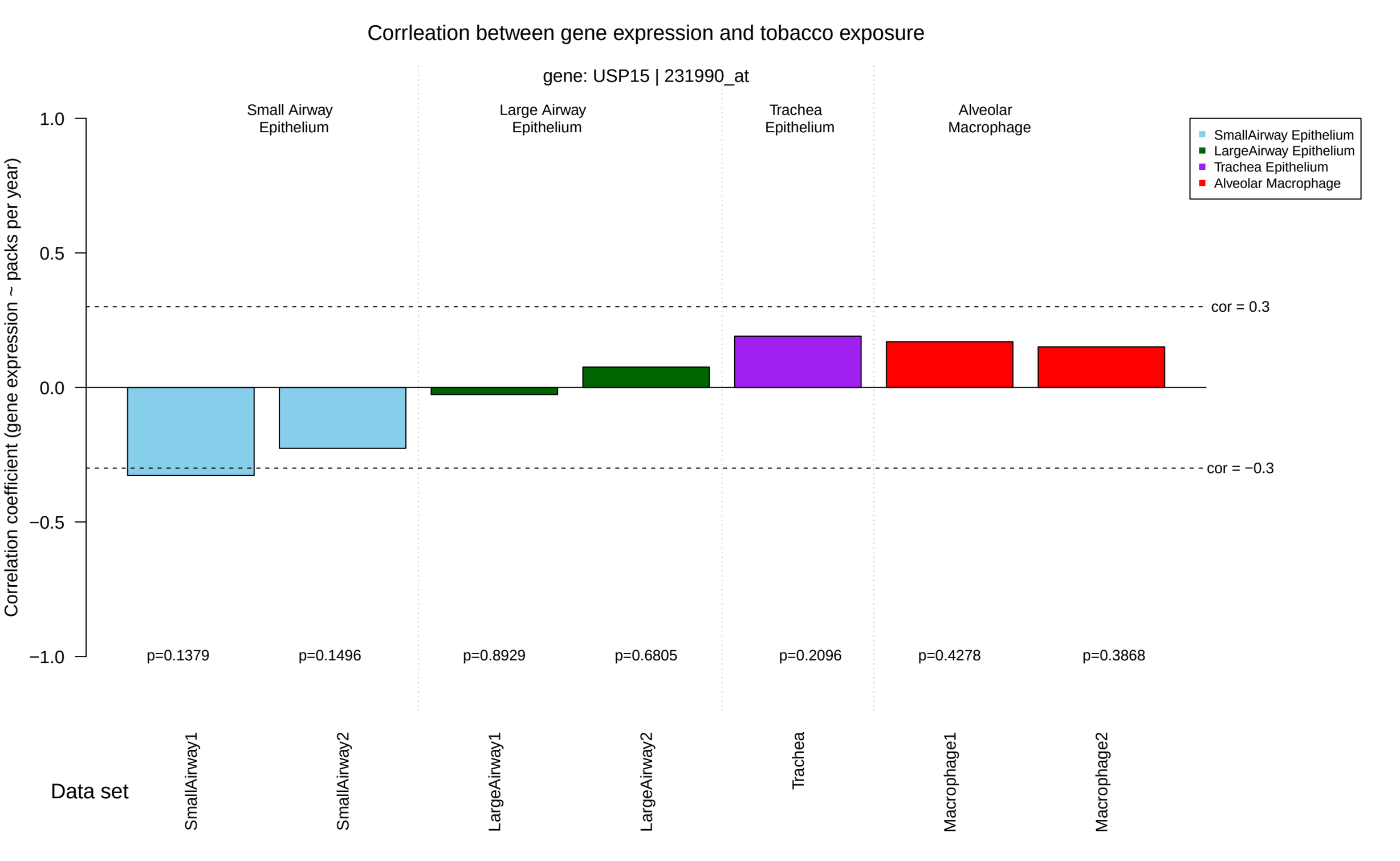

set: LargeAirway1 (p=0.9971)
USP15 | 231990_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9157)
USP15 | 231990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6461)
USP15 | 231990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4009)
USP15 | 231990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2967)
USP15 | 231990_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP15 | 231990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1379
p=0.1496
p=0.8929
p=0.6805
p=0.2096
p=0.4278
p=0.3868
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2